

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Specimen 4WM2H Worked Solutions

Specimen | 4WM2H | Unit 2

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 1** Here are five cards, where x represents a number.

15	7	-2	23	x
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The mean of the five numbers is 12

Work out the value of x

$x =$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 2 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

.....
(2)

- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

.....
(2)

.....
(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

3 $\frac{2^x}{4^n} = 2^x$

Find an expression for x in terms of k and n

$x = \dots\dots\dots$

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 2

TOPIC: **Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

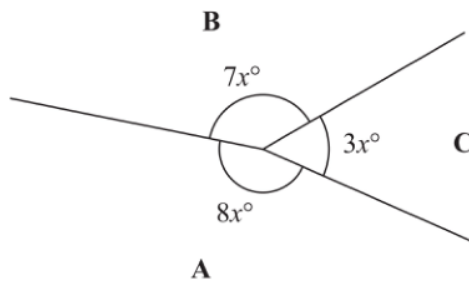


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.
Work out the value of n

$n =$

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

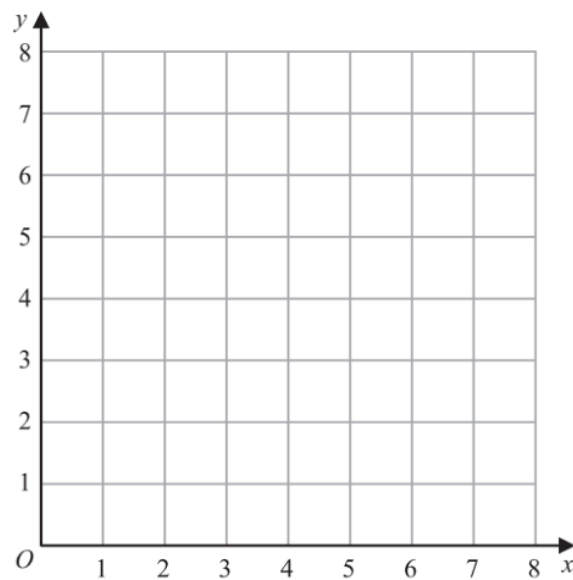
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PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

5 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$ $x \geq 2$ $x + y \leq 7$

Label the region **R**

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 4

TOPIC: **Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 6****Marks: 3**TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

6 Here are some integers where $a < b < c < d$

a b c d d d

The mode of the integers is 9

The range of the integers is 4

The median of the integers is 8

Work out the value of a , the value of b , the value of c and the value of d

$a =$

$b =$

$c =$

$d =$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 7 A cylinder is placed on the ground.

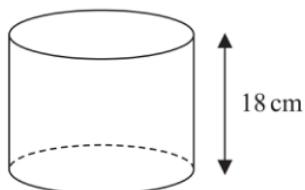


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 4

TOPIC: **Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 8****Marks: 6****TOPIC: Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 8** In 2021, the value of Asha's apartment was 634 400 euros.
The value of Asha's apartment had increased by 4% from its value in 2020

(a) Work out the value of Asha's apartment in 2020

..... euros
(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %
(3)

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 6

TOPIC: **Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 9****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

9 (a) Write 0.000089 in standard form.

.....
(1)

(b) Write 8.34×10^4 as an ordinary number.

.....
(1)

(Total for Question 9 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

10 Payel makes 300 celebration cards so that

number of number of number of
 birthday cards : anniversary cards : congratulations cards = 7 : 5 : 3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.
Give your answer in its simplest form.

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

11 Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 3

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Percentages**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

12 Zimo is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Zimo pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{2}{5}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$405

Work out how much Zimo has to pay for Payment 2

\$.....

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 5

TOPIC: Percentages

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

13 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6}$$

(a) Find $f(10)$
(1)(b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \dots$

$$f^{-1} : x \mapsto \dots$$

(3)

PAST PAPER SOLUTION

Q#: 13

Marks: 4

TOPIC: **Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

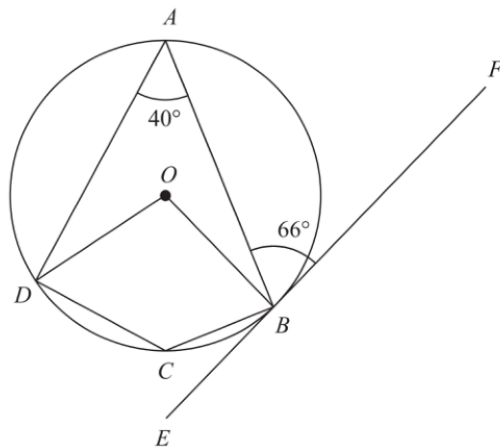
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

14 A, B, C and D are points on a circle, centre O EBF is the tangent to the circle at B Diagram **NOT**
accurately drawn(a) (i) Work out the size of angle DCB
(1)

(ii) Give a reason for your answer to (a)(i)

.....
(1)(b) Work out the size of angle ADO
(3)**(Total for Question 14 is 5 marks)**

PAST PAPER SOLUTION

Q#: 14

Marks: 5

TOPIC: **Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION

Q#: 15 Marks: 7

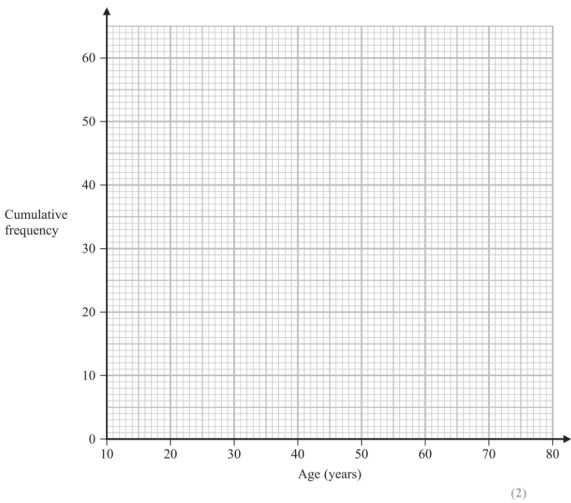
TOPIC: Cumulative Frequency Diagrams

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

15 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leq 20$	13
$10 < a \leq 30$	36
$10 < a \leq 40$	42
$10 < a \leq 50$	47
$10 < a \leq 60$	52
$10 < a \leq 70$	56
$10 < a \leq 80$	60

(a) On the grid, draw a cumulative frequency graph for the information in the table.



(b) Use your graph to find an estimate for the median of the ages of these people.

_____ years
(1)

(c) Use your graph to find an estimate for the interquartile range of the ages of these people.

_____ years
(2)

(d) Use your graph to find an estimate for the number of these people who are older than 55 years.

_____ (2)

(Total for Question 15 is 7 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 16** M is directly proportional to h^3
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

$h = \dots\dots\dots$

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 4

TOPIC: **Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 17** A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 5

TOPIC: **Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 18** Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

.....
(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18

Marks: 3

TOPIC: **Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

19 The diagram shows two similar metal statues.**A****B**Diagram **NOT**
accurately drawnThe volume of statue **B** is 20% less than the volume of statue **A**The surface area of statue **B** is $k\%$ less than the surface area of statue **A**Work out the value of k

Give your answer correct to 3 significant figures.

 $k = \dots\dots\dots$ **(Total for Question 19 is 4 marks)**

PAST PAPER SOLUTION

Q#: 19

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

$$21 \quad a = \frac{14}{3x-7} \quad x = \frac{1}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p , q , r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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EA

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

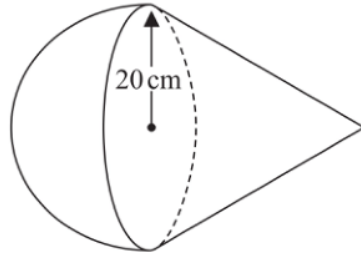
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

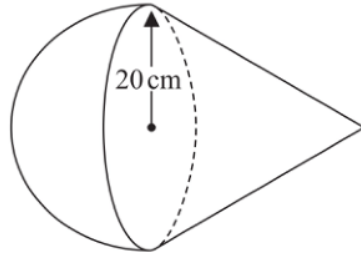
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EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

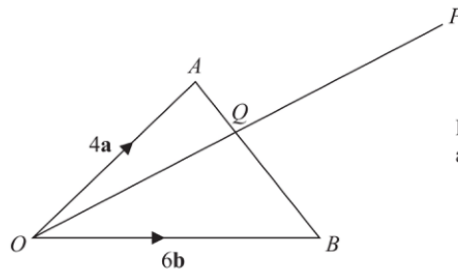
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

23Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 6\mathbf{b}$$

$$\vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

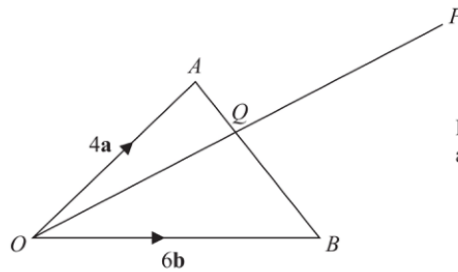
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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 23****Marks: 5**TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

23Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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No worked solution is saved yet.

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